Equating the constant term on both sides of (1),we get

$$A - B - C = 0$$

$$\Rightarrow B = A - C$$

$$\therefore \quad B = -\frac{1}{4}$$

Example 2.5

Resolve into partial fraction

$$\frac{x+1}{(x-2)^2(x+3)}$$

Solution

Let $\frac{x+1}{(x-2)^2(x+3)}$

$$= \frac{A}{(x-2)} + \frac{B}{(x-2)^2} + \frac{C}{(x+3)} \qquad \ldots(1)$$

Multiplying both the sides by $(x-2)^2(x+3)$, we get

$$x+1 = A(x-2)(x+3) + B(x+3) + C(x-2)^2 \qquad \ldots(2)$$

Put $x = 2$ in (2), we have

$$2+1 = A(0) + B(5) + C(0)$$

$$\therefore \quad B = \frac{3}{5}$$

Put $x = -3$ in (2), we have

$$-3+1 = A(0) + B(0) + C(-3-2)^2$$

$$\therefore \quad C = \frac{-2}{25}$$

Equating the coefficient of x^2 on both the sides of (2), we get

$$0 = A + C$$

$$A = -C$$

$$\therefore \quad A = \frac{2}{25}$$

Substituting the values of A, B and C in (1),

$$\frac{x+1}{(x-2)^2(x+3)} = \frac{2}{25(x-2)} + \frac{3}{5(x-2)^2} - \frac{2}{25(x+3)}$$

Example 2.6

Resolve into partial fraction

$$\frac{9}{(x-1)(x+2)^2}$$

Solution

$$\frac{9}{(x-1)(x+2)^2} = \frac{A}{(x-1)} + \frac{B}{(x+2)} + \frac{C}{(x+2)^2} \ldots (1)$$

Multiplying both sides by $(x-1)(x+2)^2$

$$\therefore 9 = A(x+2)^2 + B(x-1)(x+2) + C(x-1) \qquad \ldots (2)$$

Put $x = -2$ in (2)

$$9 = A(0) + B(0) + C(-2-1)$$

$$\therefore \quad C = -3$$

Put $x = 1$ in (2)

$$9 = A(3)^2 + B(0) + C(0)$$

$$\therefore \quad A = 1$$

Equating the coefficient of x^2 on both the sides of (2), we get

$$0 = A + B$$

$$B = -A$$

$$\therefore \quad B = -1$$

Substituting the values of A, B and C in (1), we get

$$\frac{9}{(x-1)(x+2)^2} = \frac{1}{x-1} - \frac{1}{x+2} - \frac{3}{(x+2)^2}$$

2.1.3 Denominator contains quadratic factor, which cannot be factorized into linear factors

In the rational fraction $\frac{p(x)}{q(x)}$, if one of the quadratic factor $q(x)$ is of the form $ax^2 + bx + c$ which cannot be factorized into linear factors, then $\frac{p(x)}{q(x)}$ can be

02_11th_BM-STAT_Ch-2-EM.indd 28 21-04-2020 12:18:13 PM